Assignment 2: Probability and random variables

CE/EL/MC/CV/GL/GM/ES 361 Probability and Statistics • First Semester, 2026/2027

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Chapter 2, Introduction to Probability • Chapter 3, Random Variables and Distributions

What to hand in, and when

Handwritten, in your own hand, on at most two sides of paper. Put your **name, index number and programme** at the top of the first page, and your **assigned values** (see below) immediately underneath. Nothing typed is accepted.

Show your working. A bare answer with no steps earns nothing, even when the number is right. Leave answers as exact fractions where they are exact, and give decimals to four places where they are not.

There are **two steps, and both are required**:

1. **Online, by 23:59 GMT on Wednesday 11 November 2026.** Photograph or scan your finished pages and upload them here:
   `https://docs.google.com/forms/d/e/1FAIpQLSfGAMoYCY1BY0a9GF4viSF_76DDzs-bgzRGowwJfIl-fK5-XQ/viewform`
   This is the same upload form as Assignment 1. Sign in with your university account, and say clearly in the form that this is Assignment 2. The upload is time-stamped, so late uploads are visibly late. Make sure the writing is legible in the photograph.
2. **On paper, at the first class after the deadline.** Hand in the original sheets. They must be the same pages you uploaded.

An upload with no paper, or paper with no upload, is an incomplete submission. Nothing is accepted by email or WhatsApp.

Your numbers depend on your index number

Let d be the **last digit of your index number**. Every question below uses d. Work out your four assigned values first and write them at the top of your first page, with the arithmetic shown.

Question	Symbol	Rule	Your value
Q1	r	$r = 3 + (d \bmod 3)$	crew size
Q2	p	$p = (1 + d)/100$	prevalence
Q3	a	$a = 1 + (d \bmod 3)$	shift in the mass function
Q4	b	$b = 2 + (d \bmod 3)$	upper limit of the density

Here $d \bmod 3$ means the remainder when d is divided by 3. For example, an index number ending in 7 gives $d = 7$, so $7 \bmod 3 = 1$ and therefore $r = 4$, $p = 0.08$, $a = 2$ and $b = 3$.

A script that does not match the writer's index number scores zero, whoever wrote it. Check your four values before you start: everything after this depends on them.

The questions

Q1. Counting, and a probability from it.

A maintenance crew of r people is to be chosen from a pool of 6 mechanical and 5 electrical technicians. All 11 are equally likely to be picked and nobody serves twice.

(a) How many different crews are possible? State clearly whether order matters, and why.

(b) How many of those crews contain at least one electrical technician? Do it by the complement, and say in one sentence why the complement is easier here.

(c) Find the probability that a crew chosen at random is made up entirely of mechanical technicians.

(d) The foreman wants "at least one electrical technician" guaranteed, not left to chance. Explain in two sentences why your answer to (c) is not zero, and what he would have to change about the selection to make it zero.

Q2. Conditional probability and Bayes' rule.

A rapid field test is used on borehole water to screen for excess fluoride. In the district concerned, a proportion p of boreholes genuinely exceed the limit. The test is positive for 95% of boreholes that genuinely exceed it, and is also positive for 8% of boreholes that do not.

(a) Draw a tree diagram for the two stages, with all four end probabilities.
(b) Find the probability that a randomly chosen borehole tests positive.
(c) A borehole has just tested positive. Find the probability that it genuinely exceeds the limit. Give your answer to four decimal places.
(d) Your answer to (c) is a long way below 95%. Explain to a district water officer, in plain language and without algebra, why that is so.
(e) Classmates with different index numbers will get different answers to (c). Say which way the answer moves as p rises, and why that is what you would expect.

Q3. A discrete random variable.

Let X be the number of haul trucks out of service at the start of a shift. Its probability mass function is

$$p(x) \;=\; k\,(x + a), \qquad x = 0,\ 1,\ 2,\ 3,$$

and $p(x) = 0$ elsewhere, where a is your assigned value.

(a) Find k. Leave it as an exact fraction.
(b) Present the mass function as a table and verify that it sums to 1.
(c) Find $P(X \geq 2)$.
(d) Write down the cumulative distribution function $F(x)$ for all real x, and sketch it.
(e) Your sketch is a staircase. State what the height of each step represents, and why the function is flat between the steps.

Q4. A continuous random variable.

A jackhammer is serviced and returned to the face. Let X be the time in hours until its first stoppage, with probability density function

$$f(x) \;=\; c\,(b - x), \qquad 0 \leq x \leq b,$$

and $f(x) = 0$ elsewhere, where b is your assigned value.

(a) Find c. Leave it as an exact fraction.
(b) Sketch the density. The curve falls as x grows: say what that means about when stoppages happen, in one sentence an operator would understand.
(c) Derive the cumulative distribution function $F(x)$ for $0 \leq x \leq b$, and check that $F(b) = 1$.
(d) Find $P(X > 1)$, the probability that the machine runs for more than one hour.
(e) Find the median time to first stoppage. You will need to solve a quadratic; keep the root that lies inside $[0, b]$ and say why the other one is rejected.

Honesty: read this part carefully

This is an **individual, handwritten** assignment. You may discuss ideas with classmates and you may read anything you like, but the script you hand in must be written by you, in your own hand, in your own words.

Artificial intelligence tools must not be used to write any part of it. You may use them to understand the topic, in the same way you would use a textbook, but work generated by a tool and copied onto your page is not your work and will be treated as such.

Copying another student's script means both of you receive zero, the copier and the one who supplied it.

How this is checked. Three ways, all simple. First, every script is specific to your index number: r, p, a and b all differ between students, so a shared or generated answer usually solves the wrong problem,

and the numbers give it away at a glance. Second, the script you hand in on paper is compared with the photograph you uploaded, and the upload is time-stamped. Third, a random selection of students will be asked, in class, to **reproduce one part of their own script at the board**, with their assigned values, without looking at it. Students who cannot do that for work bearing their name will have the submission disallowed.

I am not using AI-detection software, because it is unreliable and unfair to students writing in a second language. I am relying on the three checks above, which are not.

Copy this declaration at the bottom of your last page and sign it:

This script is my own handwriting and my own words. I did not copy it from another student or from any source, and I did not use an AI tool to write it.

Signature	Index number	Date

Everything here is covered in Chapters 2 and 3 of the lecture notes, on the course page, and practised in Tutorials 2 and 3. Bring questions to the next class rather than guessing: that is what the class is for.